

URL SHORTENER - SYSTEM DESIGN

Requirements, Assumptions and Capacity Planning

FUNCTIONAL REQUIREMENTS

- Convert a long URL into a short URL
- Redirect users from the short URL to the original URL
- Allow optional custom aliases
- Support link expiration
- Count clicks and collect basic analytics
- Allow registered users to view and delete their URLs

NON-FUNCTIONAL REQUIREMENTS

- Availability target: 99.99% for redirects
- Redirect latency target: below 100 ms
- Horizontally scalable services
- Read-heavy workload
- Short codes should be compact and difficult to guess
- URL mappings must be durable
- Analytics must not slow down redirects

CAPACITY ESTIMATION

TRAFFIC ASSUMPTIONS

New URLs = 50 million / month
Read : Write = 100 : 1

Write QPS $\approx$ 19 URLs/sec
Read QPS $\approx$ 1,900 redirects/sec

Peak factor = 3x
Peak read traffic $\approx$ 5,700 requests/sec

STORAGE

Retention = 5 years

$50M \times 12 \times 5 = 3$ billion URLs

Estimated record size $\approx$ 700 bytes

Raw storage $\approx$ 2.1 TB
With 3x replication $\approx$ 6.3 TB

CACHE & BANDWIDTH

Assume 80% traffic comes from 20% URLs

Hot URLs are stored in Redis

Average request payload $\approx$ 700 bytes

Redis keeps frequently accessed mappings to reduce database load and latency

DESIGN DECISIONS

Stateless application servers $\rightarrow$ horizontal scaling

Redis $\rightarrow$ fast URL lookup

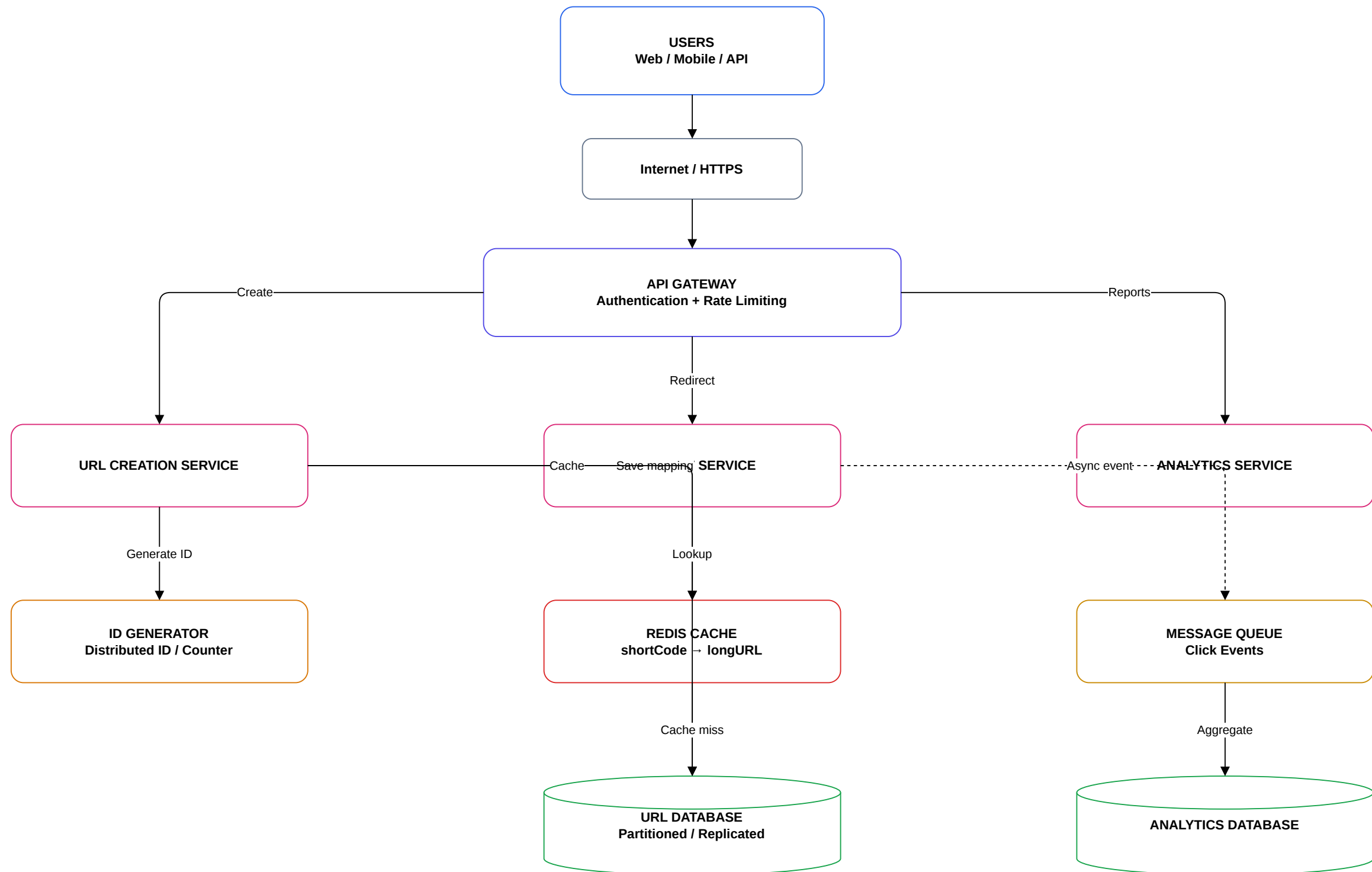
Sharded database $\rightarrow$ large-scale storage

Base62 $\rightarrow$ compact short codes

Message queue $\rightarrow$ asynchronous click processing

Load balancer $\rightarrow$ distribute incoming requests

URL SHORTENER - HIGH LEVEL ARCHITECTURE



REQUEST FLOW AND REST API DESIGN

A. CREATE SHORT URL



B. REDIRECT FLOW



REST API ENDPOINTS

POST /api/v1/urls

Request: { longUrl, customAlias?, expiresAt? }

Response: 201 Created
Returns shortCode and shortUrl

GET /{shortCode}

Returns HTTP 302 redirect

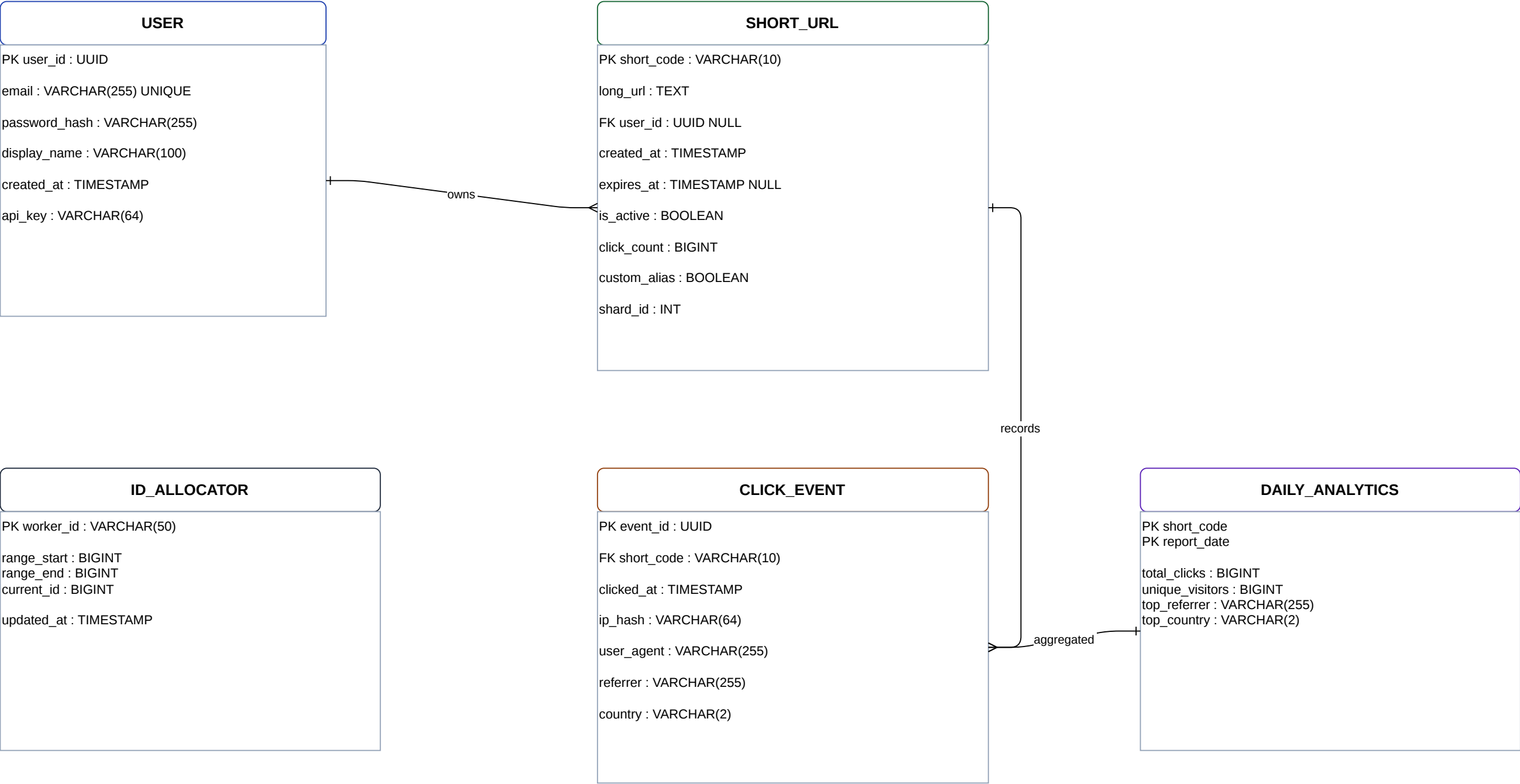
404 = code not found
410 = link expired

GET /api/v1/urls/{code}/analytics

Returns clicks by day, referrers and country information

Authenticated endpoint

URL SHORTENER - DATABASE MODEL



Design note: anonymous users may create URLs, therefore user_id can be NULL.